

PCB Artwork Design

EOM MIN HO

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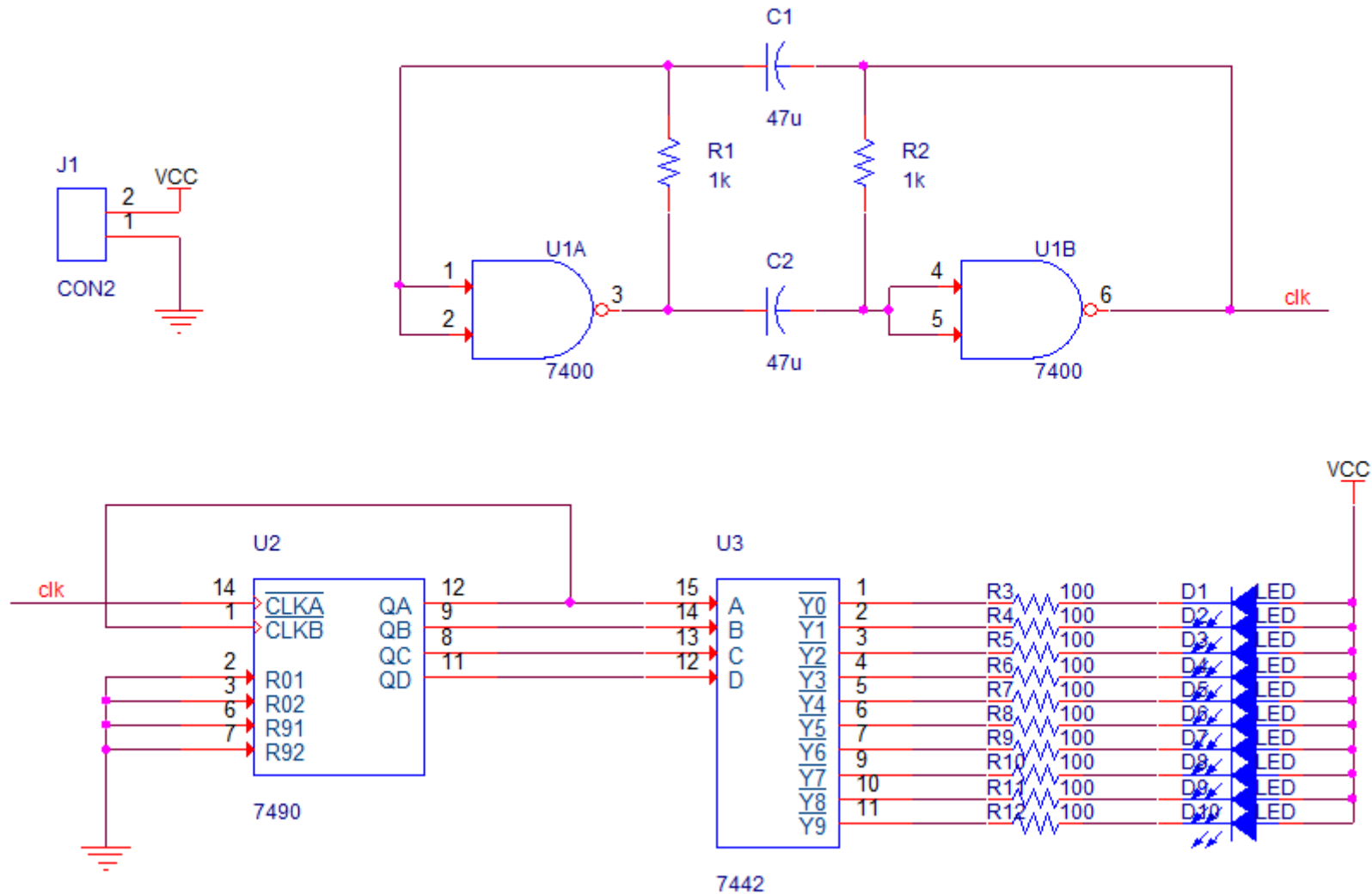
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1. Project Overview

Decade Counter LED Sequencer – Schematic Diagram



Parts List

No.	Part Name (Capture)	Value	Qty	Library
1	CON2	2-pin	1	Connector.olb
2	C	47 μ F	2	Discrete.olb
3	R	1 k Ω	2	analog.olb
		100 Ω	10	
4	7400	NAND Gate	2	eval.olb
5	7490	Decade Counter	1	Counter.olb
6	7442	BCD to Decimal Decoder	1	MuxDecoder.olb
7	D1–D10	LED (Red)	10	Discrete.olb

Component Footprint

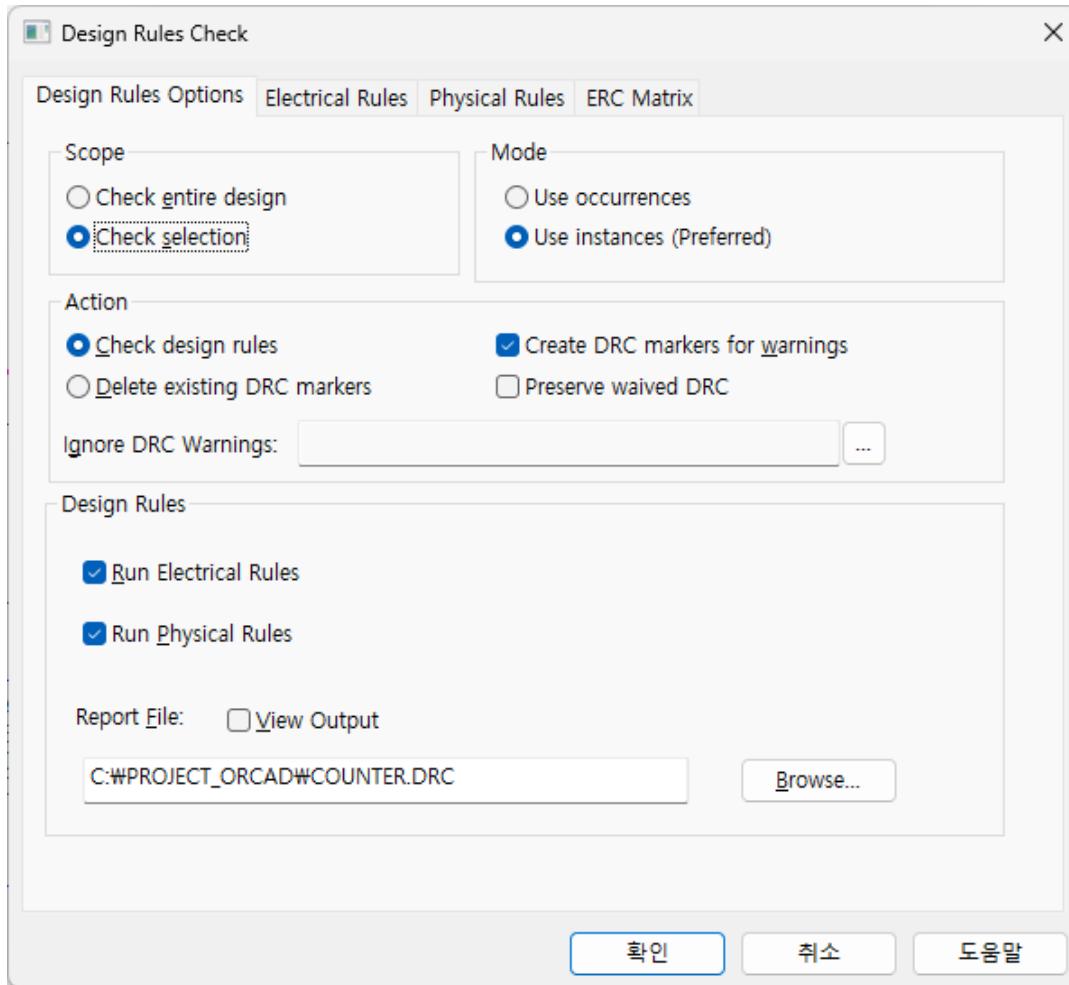
Property Editor

New Property... Apply Display... Delete Property Pivot Filter by: < Current properties > Help

		Part Reference	PCB Footprint	POWER	Power Pins Visible	Primitive
1	+	SCHEMATIC1 : PAGE1	C1	cap196	☐	DEFAULT
2	+	SCHEMATIC1 : PAGE1	C2	cap196	☐	DEFAULT
3	+	SCHEMATIC1 : PAGE1	D1	cap196	☐	DEFAULT
4	+	SCHEMATIC1 : PAGE1	D2	cap196	☐	DEFAULT
5	+	SCHEMATIC1 : PAGE1	D3	cap196	☐	DEFAULT
6	+	SCHEMATIC1 : PAGE1	D4	cap196	☐	DEFAULT
7	+	SCHEMATIC1 : PAGE1	D5	cap196	☐	DEFAULT
8	+	SCHEMATIC1 : PAGE1	D6	cap196	☐	DEFAULT
9	+	SCHEMATIC1 : PAGE1	D7	cap196	☐	DEFAULT
10	+	SCHEMATIC1 : PAGE1	D8	cap196	☐	DEFAULT
11	+	SCHEMATIC1 : PAGE1	D9	cap196	☐	DEFAULT
12	+	SCHEMATIC1 : PAGE1	D10	cap196	☐	DEFAULT
13	+	SCHEMATIC1 : PAGE1	J1	jumper2	☐	DEFAULT
14	+	SCHEMATIC1 : PAGE1	R1	res400	☐	DEFAULT
15	+	SCHEMATIC1 : PAGE1	R2	res400	☐	DEFAULT
16	+	SCHEMATIC1 : PAGE1	R3	res400	☐	DEFAULT
17	+	SCHEMATIC1 : PAGE1	R4	res400	☐	DEFAULT
18	+	SCHEMATIC1 : PAGE1	R5	res400	☐	DEFAULT
19	+	SCHEMATIC1 : PAGE1	R6	res400	☐	DEFAULT
20	+	SCHEMATIC1 : PAGE1	R7	res400	☐	DEFAULT
21	+	SCHEMATIC1 : PAGE1	R8	res400	☐	DEFAULT
22	+	SCHEMATIC1 : PAGE1	R9	res400	☐	DEFAULT
23	+	SCHEMATIC1 : PAGE1	R10	res400	☐	DEFAULT
24	+	SCHEMATIC1 : PAGE1	R11	res400	☐	DEFAULT
25	+	SCHEMATIC1 : PAGE1	R12	res400	☐	DEFAULT
26	+	SCHEMATIC1 : PAGE1	U1A	dip14_3	☐	DEFAULT
27	+	SCHEMATIC1 : PAGE1	U1B	dip14_3	☐	DEFAULT
28	+	SCHEMATIC1 : PAGE1	U2	dip14_3	☐	DEFAULT
29	+	SCHEMATIC1 : PAGE1	U3	dip16_3	☐	DEFAULT

Parts Schematic Nets Flat Nets Pins Title Blocks Globals Ports Aliases

Design Rule Check (DRC)



The Design Rules Check dialog box is shown with the 'Design Rules Options' tab selected. It contains sections for Scope, Mode, Action, Ignore DRC Warnings, and Design Rules. The 'Check selection' option is selected under Scope, and 'Use instances (Preferred)' is selected under Mode. Under Action, 'Check design rules' and 'Create DRC markers for warnings' are selected. The 'Design Rules' section has 'Run Electrical Rules' and 'Run Physical Rules' checked. The report file is set to 'C:\PROJECT_ORCAD\COUNTER.DRC'.

Design Rules Check

Design Rules Options | Electrical Rules | Physical Rules | ERC Matrix

Scope

☐ Check entire design

☒ Check selection

Mode

☐ Use occurrences

☒ Use instances (Preferred)

Action

☒ Check design rules

☐ Delete existing DRC markers

☒ Create DRC markers for warnings

☐ Preserve waived DRC

Ignore DRC Warnings: ...

Design Rules

☒ Run Electrical Rules

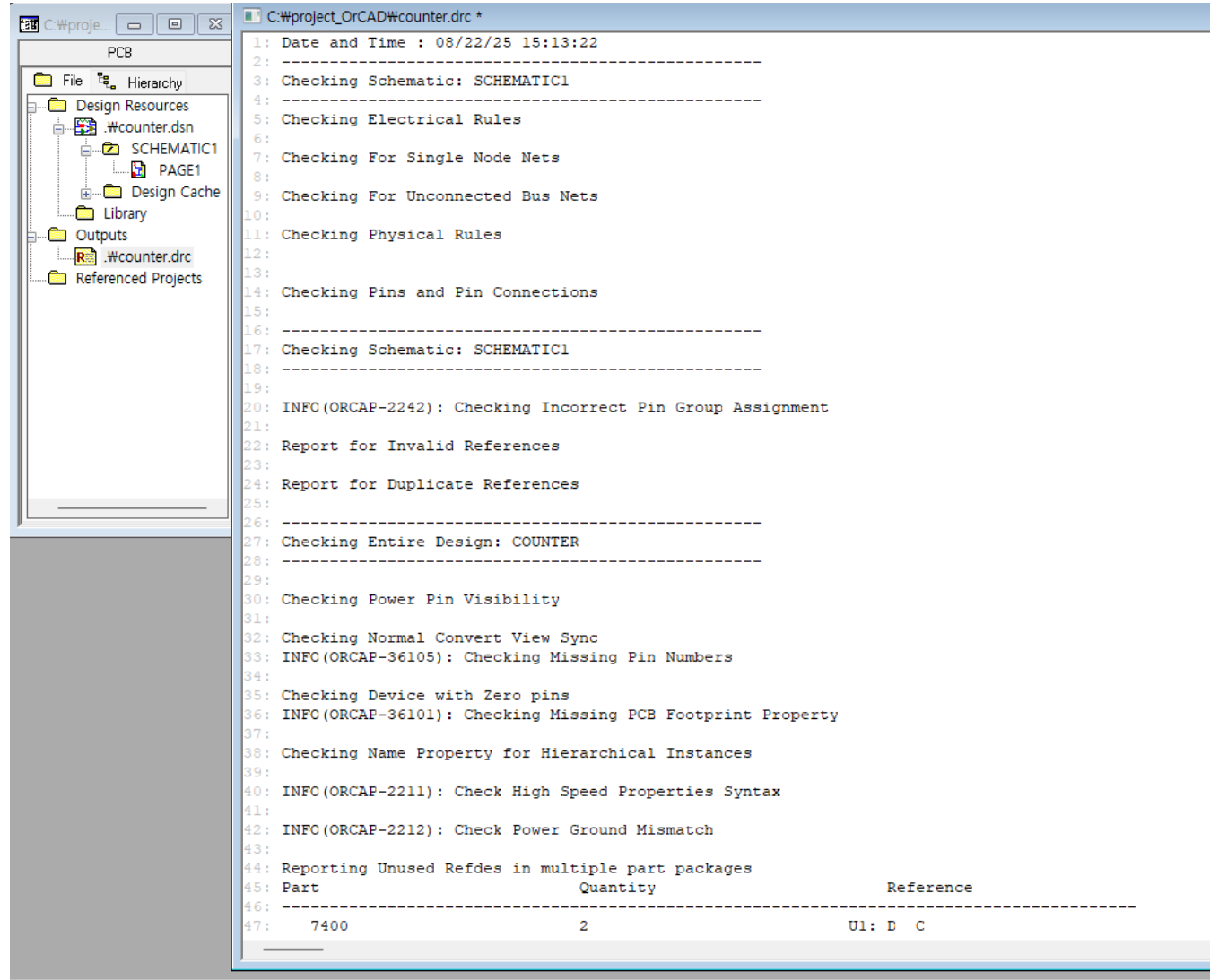
☒ Run Physical Rules

Report File: ☐ View Output

C:\PROJECT_ORCAD\COUNTER.DRC

Browse...

확인 취소 도움말



The image shows a PCB project hierarchy on the left and a DRC report on the right. The hierarchy includes Design Resources, Schematic1, Page1, Design Cache, Library, Outputs, Wcounter.drc, and Referenced Projects. The DRC report shows the progress of the check, including checking schematic, electrical rules, physical rules, and pins. It also includes a table for reporting unused references in multiple part packages.

PCB

File Hierarchy

Design Resources

.Wcounter.dsn

Schematic1

PAGE1

Design Cache

Library

Outputs

.Wcounter.drc

Referenced Projects

C:\project_OrCAD\Wcounter.drc *

```
1: Date and Time : 08/22/25 15:13:22
2: -----
3: Checking Schematic: SCHEMATIC1
4: -----
5: Checking Electrical Rules
6: -----
7: Checking For Single Node Nets
8: -----
9: Checking For Unconnected Bus Nets
10: -----
11: Checking Physical Rules
12: -----
13: -----
14: Checking Pins and Pin Connections
15: -----
16: -----
17: Checking Schematic: SCHEMATIC1
18: -----
19: -----
20: INFO(ORCAP-2242): Checking Incorrect Pin Group Assignment
21: -----
22: Report for Invalid References
23: -----
24: Report for Duplicate References
25: -----
26: -----
27: Checking Entire Design: COUNTER
28: -----
29: -----
30: Checking Power Pin Visibility
31: -----
32: Checking Normal Convert View Sync
33: INFO(ORCAP-36105): Checking Missing Pin Numbers
34: -----
35: Checking Device with Zero pins
36: INFO(ORCAP-36101): Checking Missing PCB Footprint Property
37: -----
38: Checking Name Property for Hierarchical Instances
39: -----
40: INFO(ORCAP-2211): Check High Speed Properties Syntax
41: -----
42: INFO(ORCAP-2212): Check Power Ground Mismatch
43: -----
44: Reporting Unused Refdes in multiple part packages
45: Part Quantity Reference
46: -----
47: 7400 2 U1: D C
```

Bill of Materials(BOM)

The screenshot displays a PCB design software interface. On the left, a file hierarchy tree is visible under the 'PCB' tab. The tree structure is as follows:

- PCB
 - File
 - Hierarchy
 - Design Resources
 - .#counter.dsn
 - SCHEMATIC1
 - PAGE1
 - Design Cache
 - Library
 - Outputs
 - .#counter.bom
 - .#counter.drc
 - Referenced Projects

On the right, the 'C:\project_OrCAD\#counter.bom' file is open, showing a Bill of Materials (BOM) table. The table includes a revision history at the top and a main table with columns: Item, Quantity, Reference, Part, and PCB Footprint.

Revision History:

- 1: Revised: Friday, August 22, 2025
- 2: Revision:
- 3:
- 4:
- 5:
- 6:
- 7:
- 8:
- 9:

Main BOM Table:

Item	Quantity	Reference	Part	PCB Footprint
15: 1	2	C1,C2	47u/t	
16: 2	10	D1,D2,D3,D4,D5,D6,D7,D8,	LED/t	
17:		D9,D10		
18: 3	1	J1	CON2/t	
19: 4	2	R1,R2	1k/t	
20: 5	10	R3,R4,R5,R6,R7,R8,R9,R10,	100/t	
21:		R11,R12		
22: 6	1	U1	7400/t	
23: 7	1	U2	7490/t	
24: 8	1	U3	7442/t	
25:				

Cross Reference

Cross Reference Parts

Scope

☒ Cross reference entire design

☐ Cross reference selection

Mode

☒ Use instances (Preferred)

☐ Use occurrences

Sorting

☒ Sort output by part value, then by reference designator

☐ Sort output by reference designator, then by part value

Report

☒ Report the X and Y coordinates of all parts

☐ Report unused parts in multiple part packages

Report File:

☒ Save as XRF ☐ Save as CSV

☒ View Output

C:\PROJECT_ORCAD\COUNTER.XRF

Browse...

OK

Cancel

Help

C:\project_OrCAD\counter.xrf

1: Revised: Friday, August 22, 2025

2: Revision:

3:

4:

5:

6:

7:

8:

9:

10: Design Name: C:\PROJECT_ORCAD\COUNTER.DSN

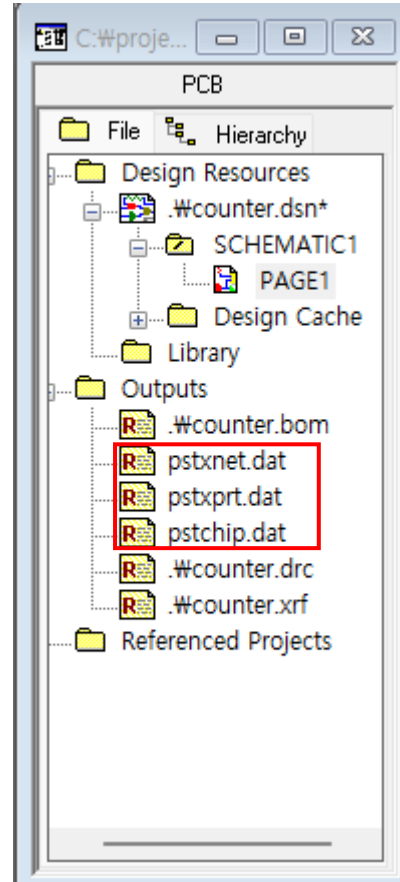
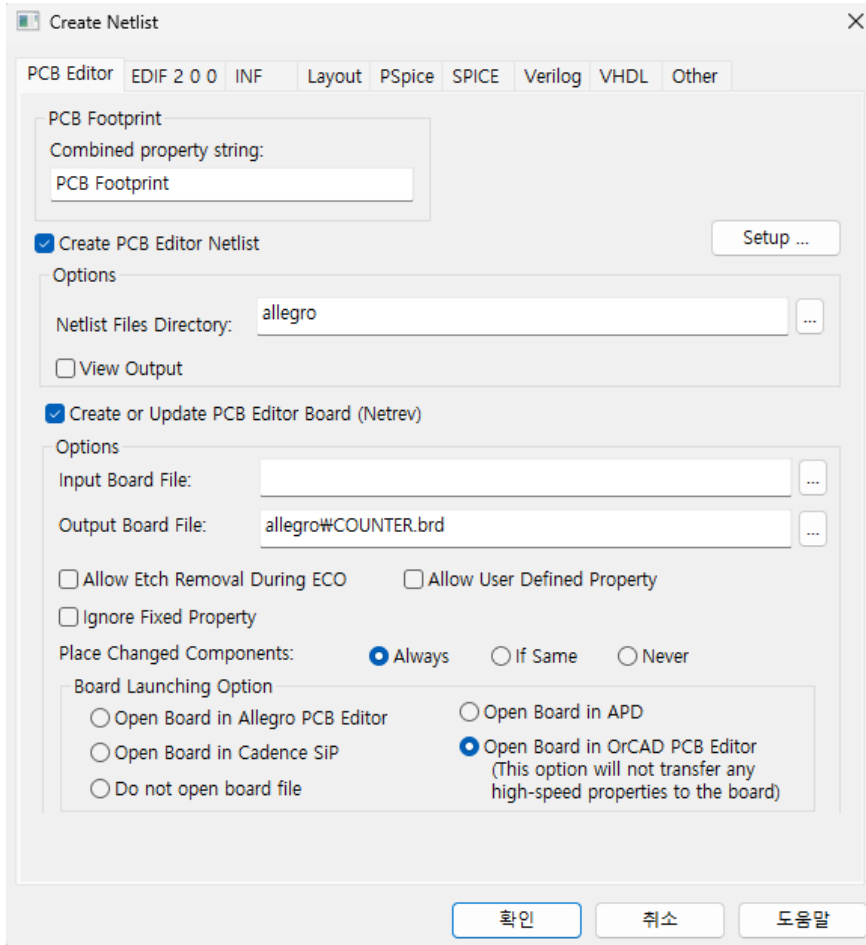
11:

12: Cross Reference August 22,2025 15:26:06 Page1

13:

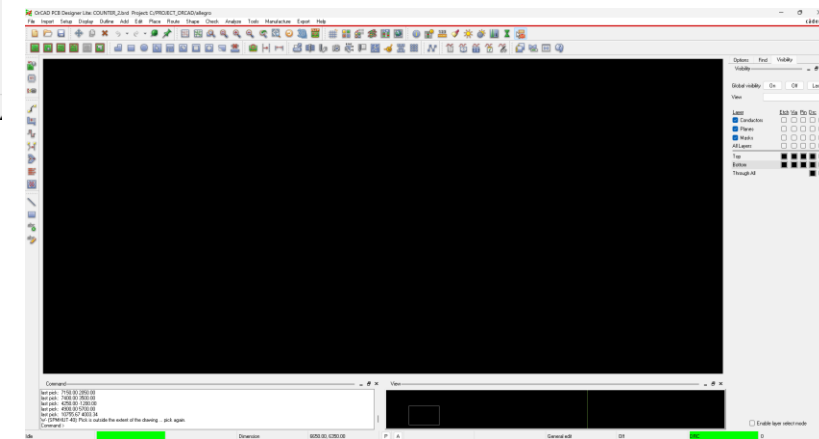
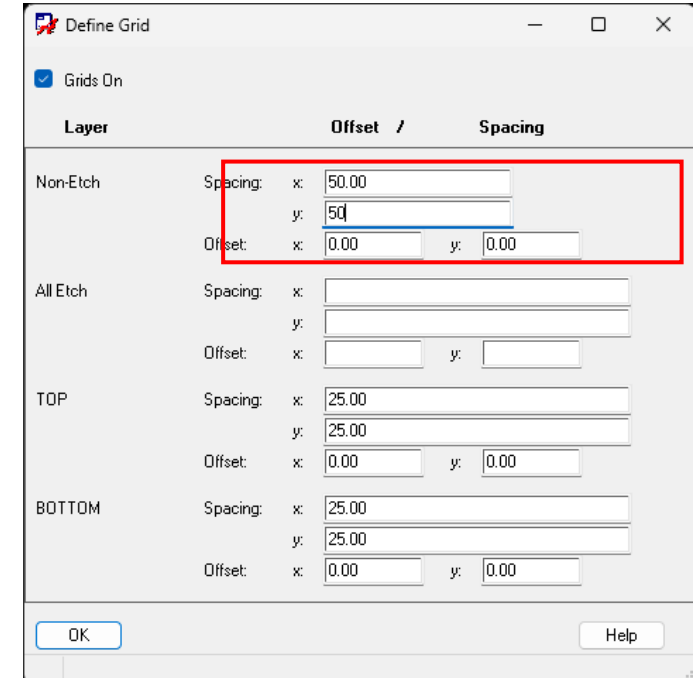
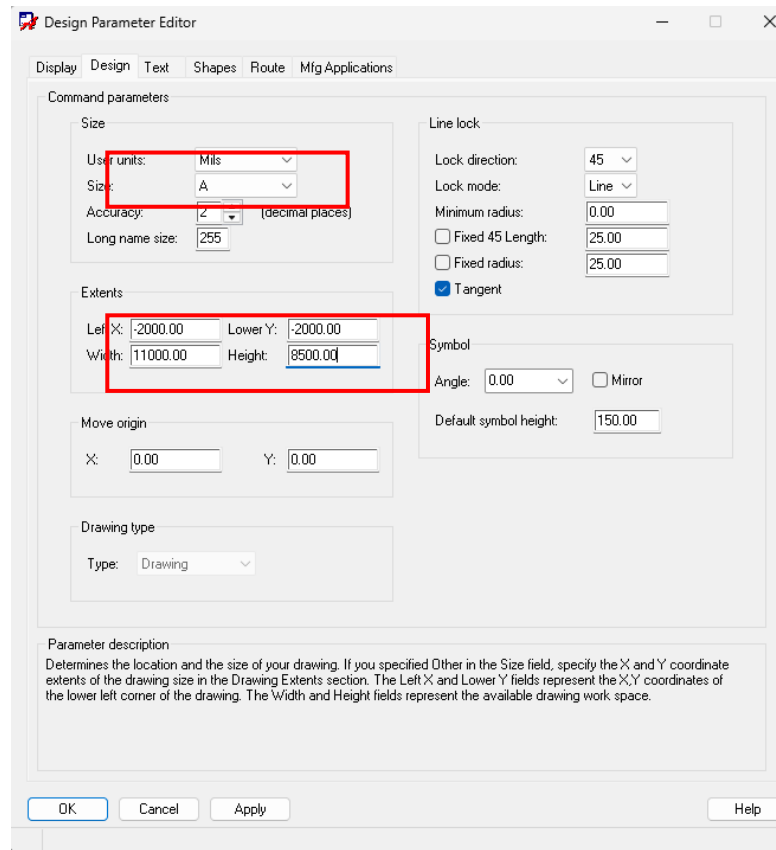
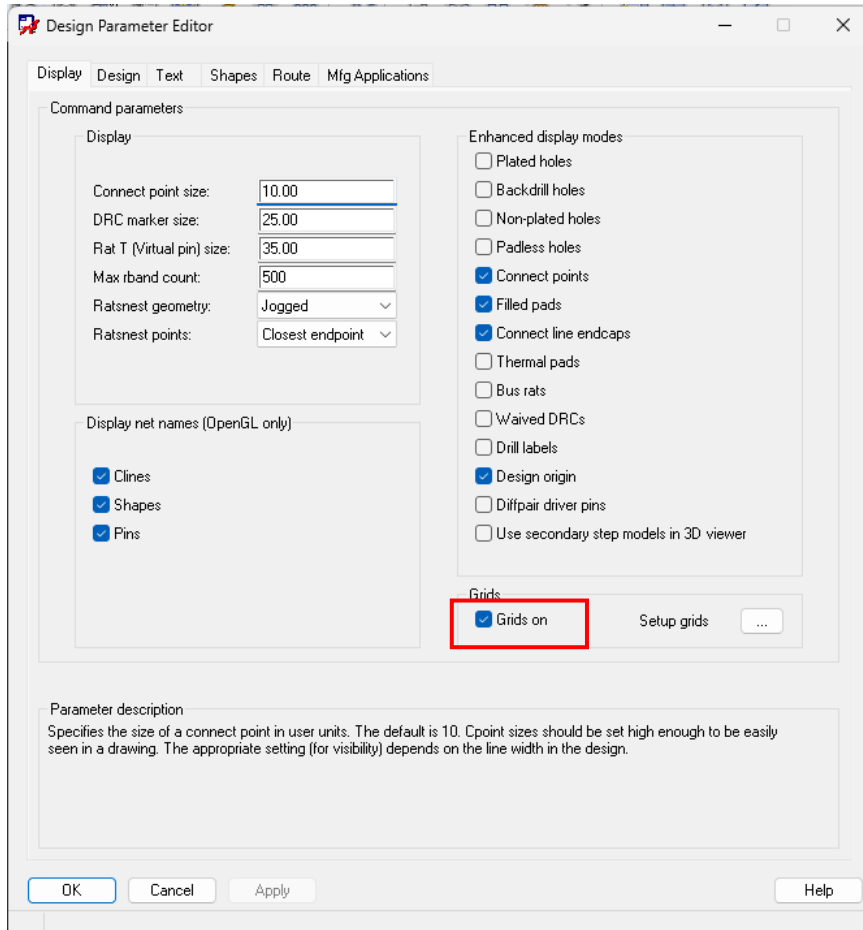
14: Item	Part	Reference	SchematicName	Sheet	Library X	Y
17: 1	1K R1	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\PSPICE\ANALOG.OLB	3.90	1.20
18: 2	1K R2	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\PSPICE\ANALOG.OLB	4.70	1.20
19: 3	47U C1	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\DISCRETE.OLB	4.40	0.80
20: 4	47U C2	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\DISCRETE.OLB	4.40	1.80
21: 5	100 R3	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\PSPICE\ANALOG.OLB	5.40	3.00
22: 6	100 R4	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\PSPICE\ANALOG.OLB	5.40	3.10
23: 7	100 R5	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\PSPICE\ANALOG.OLB	5.40	3.20
24: 8	100 R6	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\PSPICE\ANALOG.OLB	5.40	3.30
25: 9	100 R7	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\PSPICE\ANALOG.OLB	5.40	3.40
26: 10	100 R8	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\PSPICE\ANALOG.OLB	5.40	3.50
27: 11	100 R9	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\PSPICE\ANALOG.OLB	5.40	3.60
28: 12	100 R10	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\PSPICE\ANALOG.OLB	5.40	3.70
29: 13	100 R11	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\PSPICE\ANALOG.OLB	5.40	3.80
30: 14	100 R12	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\PSPICE\ANALOG.OLB	5.40	3.90
31: 15	7400 U1A	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\PSPICE\EVAL.OLB	3.20	1.70
32: 16	7400 U1B	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\PSPICE\EVAL.OLB	5.20	1.70
33: 17	7442 U3	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\MUXDECODER.OLB	4.20	3.00
34: 18	7490 U2	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\COUNTER.OLB	2.30	3.00
35: 19	CON2 J1	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\CONNECTOR.OLB	1.50	1.40
36: 20	LED D1	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\DISCRETE.OLB	6.10	3.00
37: 21	LED D2	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\DISCRETE.OLB	6.10	3.10
38: 22	LED D3	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\DISCRETE.OLB	6.10	3.20
39: 23	LED D4	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\DISCRETE.OLB	6.10	3.30
40: 24	LED D5	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\DISCRETE.OLB	6.10	3.40
41: 25	LED D6	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\DISCRETE.OLB	6.10	3.50
42: 26	LED D7	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\DISCRETE.OLB	6.10	3.60
43: 27	LED D8	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\DISCRETE.OLB	6.10	3.70
44: 28	LED D9	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\DISCRETE.OLB	6.10	3.80
45: 29	LED D10	SCHEMATIC1/PAGE1	1	C:\CADENCE\SPB_17.2\TOOLS\CAPTURE\LIBRARY\DISCRETE.OLB	6.10	3.90
46:						

Create Netlist

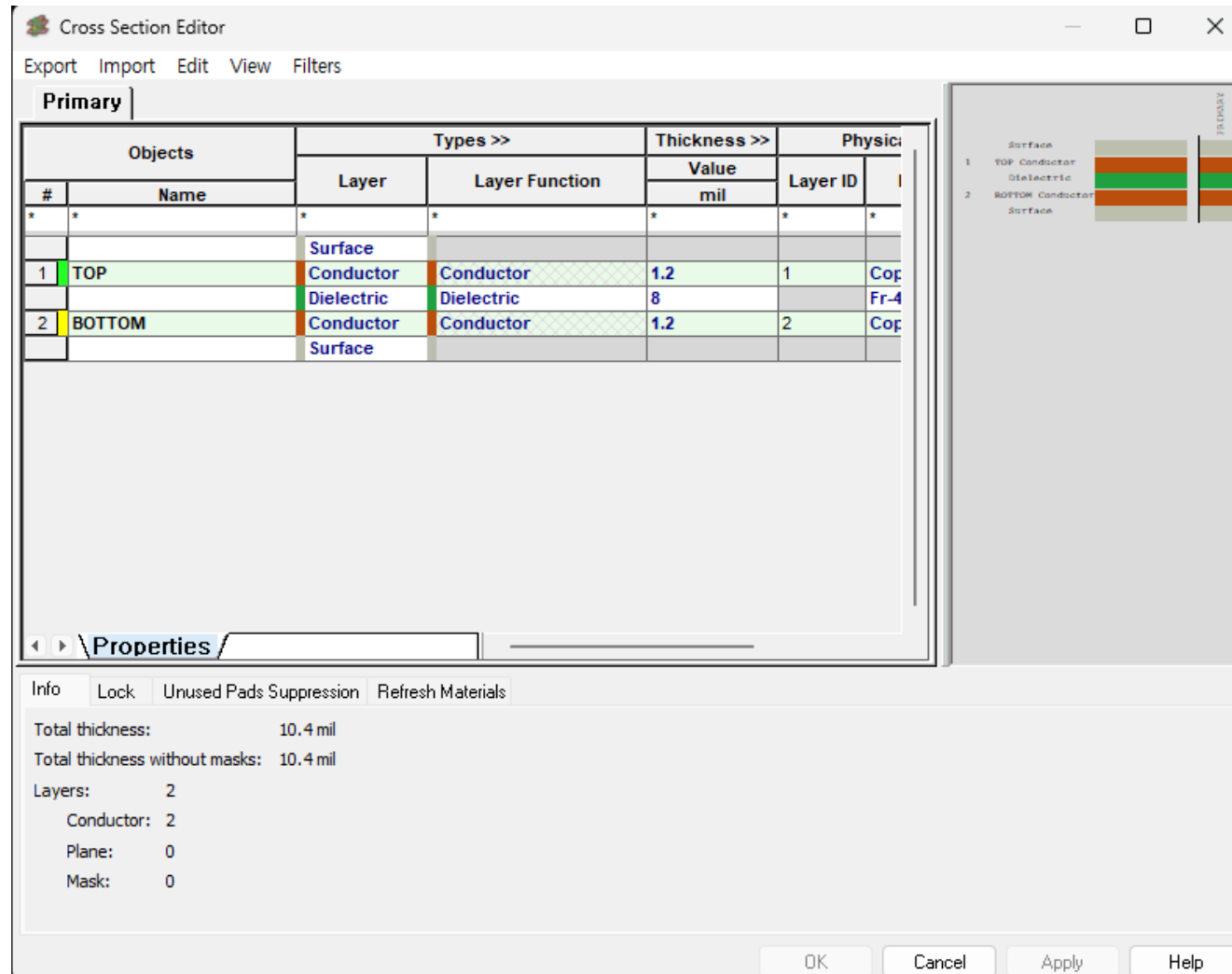


2. Pre-Layout Setup

PCB Editor Initial Settings

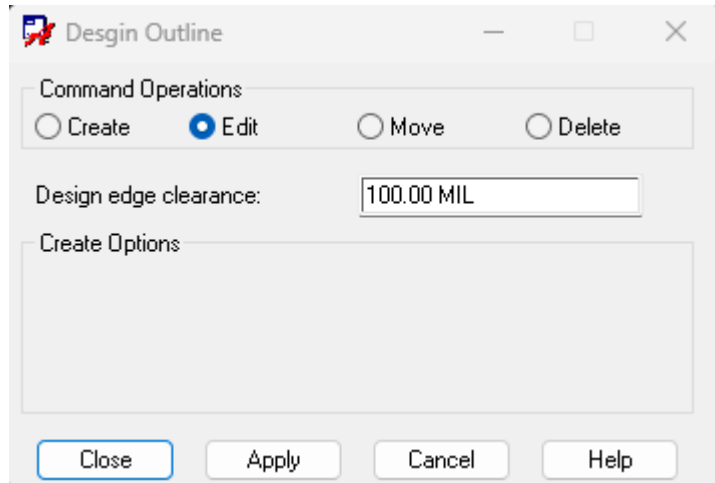
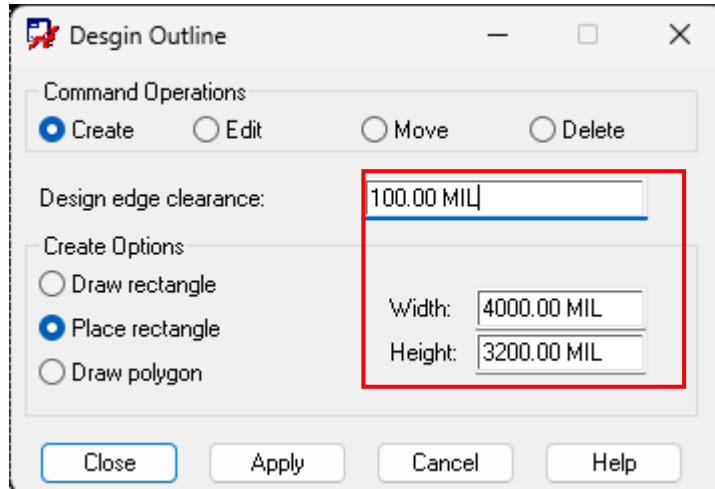


PCB Editor Initial Settings

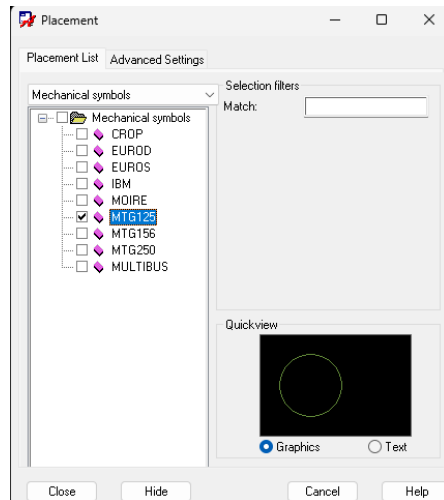
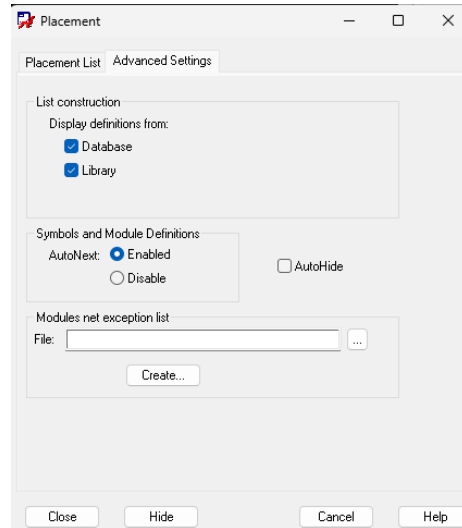


2-layer PCB

Board Outline Setup



Board Outline Setup



Constraints setup

Allegro Constraint Manager (connected to OrCAD PCB Designer Lite 17.2) [COUNTER] - [Physical / Physical Constraint Set / All Layers]

File Edit Objects Column View Analyze Audit Tools Window Help

Worksheet Selector: Electrical, Physical (selected)

Physical Constraint Set

- Net
 - All Layers
- Region
 - All Layers

Objects		Referenced Physical CSet	Line Width		Neck		Min Line Spaci mil	Primary mil
Type	Name		Min mil	Max mil	Min Width mil	Max Length mil		
Dsn	COUNTER	DEFAULT	12.00	0.00	5.00	0.00	0.00	0.00
PCS	DEFAULT		12.00	0.00	5.00	0.00	0.00	0.00

Spacing
Same Net Spacing
Properties
DRC

Idle DRC Sync on.

Constraints setup

Allegro Constraint Manager (connected to OrCAD PCB Designer Lite 17.2) [COUNTER] - [Physical / Net / All Layers]

File Edit Objects Column View Analyze Audit Tools Window Help

Worksheet Selector: COUNTER

- Electrical
- Physical
- Physical Constraint Set
 - All Layers
- Net
 - All Layers
- Region
 - All Layers

Objects		Referenced Physical C Set	Line Width		Neck		Uncoupled Length	
Type	S		Min mil	Max mil	Min Width mil	Max Length mil	Gather Control	
Dsn	COUNTER	DEFAULT	12.00	0.00	5.00	0.00		
OType	XNets/Nets							
Net	CLK	DEFAULT	12.00	0.00	5.00	0.00		
Net	GND	DEFAULT	30.00	0.00	5.00	0.00		
Net	N00248	DEFAULT	12.00	0.00	5.00	0.00		
Net	N00295	DEFAULT	12.00	0.00	5.00	0.00		
Net	N00299	DEFAULT	12.00	0.00	5.00	0.00		
Net	N00580	DEFAULT	12.00	0.00	5.00	0.00		
Net	N00584	DEFAULT	12.00	0.00	5.00	0.00		
Net	N00588	DEFAULT	12.00	0.00	5.00	0.00		
Net	N00592	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01015	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01019	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01023	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01027	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01031	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01035	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01039	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01043	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01047	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01051	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01055	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01067	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01079	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01091	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01103	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01115	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01127	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01139	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01151	DEFAULT	12.00	0.00	5.00	0.00		
Net	N01163	DEFAULT	12.00	0.00	5.00	0.00		
Net	VCC	DEFAULT	30.00	0.00	5.00	0.00		

Net: N00295

Idle DRC Sync on. XNET

Constraints setup

Allegro Constraint Manager (connected to OrCAD PCB Designer Lite 17.2) [COUNTER] - [Spacing / Spacing Constraint Set / All Layers]

File Edit Objects Column View Analyze Audit Tools Window Help

Worksheet Selector 5 X

- Electrical
- Physical
- Spacing
- Spacing Constraint Set
 - All Layers
- Net
 - All Layers
- Net Class-Class
 - All Layers
 - CSet assignment matrix
- Region
 - All Layers
- Inter Layer
 - Spacing

COUNTER

Objects		Referenced Spacing CSet	Line To >>	Thru Pin To >>	SMD Pin To >>	Test Pin To >>	Thru Via To >>	BB Via To >>
Type	S		All mil	All mil	All mil	All mil	All mil	All mil
Dsn	COUNTER	DEFAULT	10.00	10.00	5.00	5.00	10.00	5.00
SCS	DEFAULT		10.00	10.00	***	***	10.00	***

Same Net Spacing

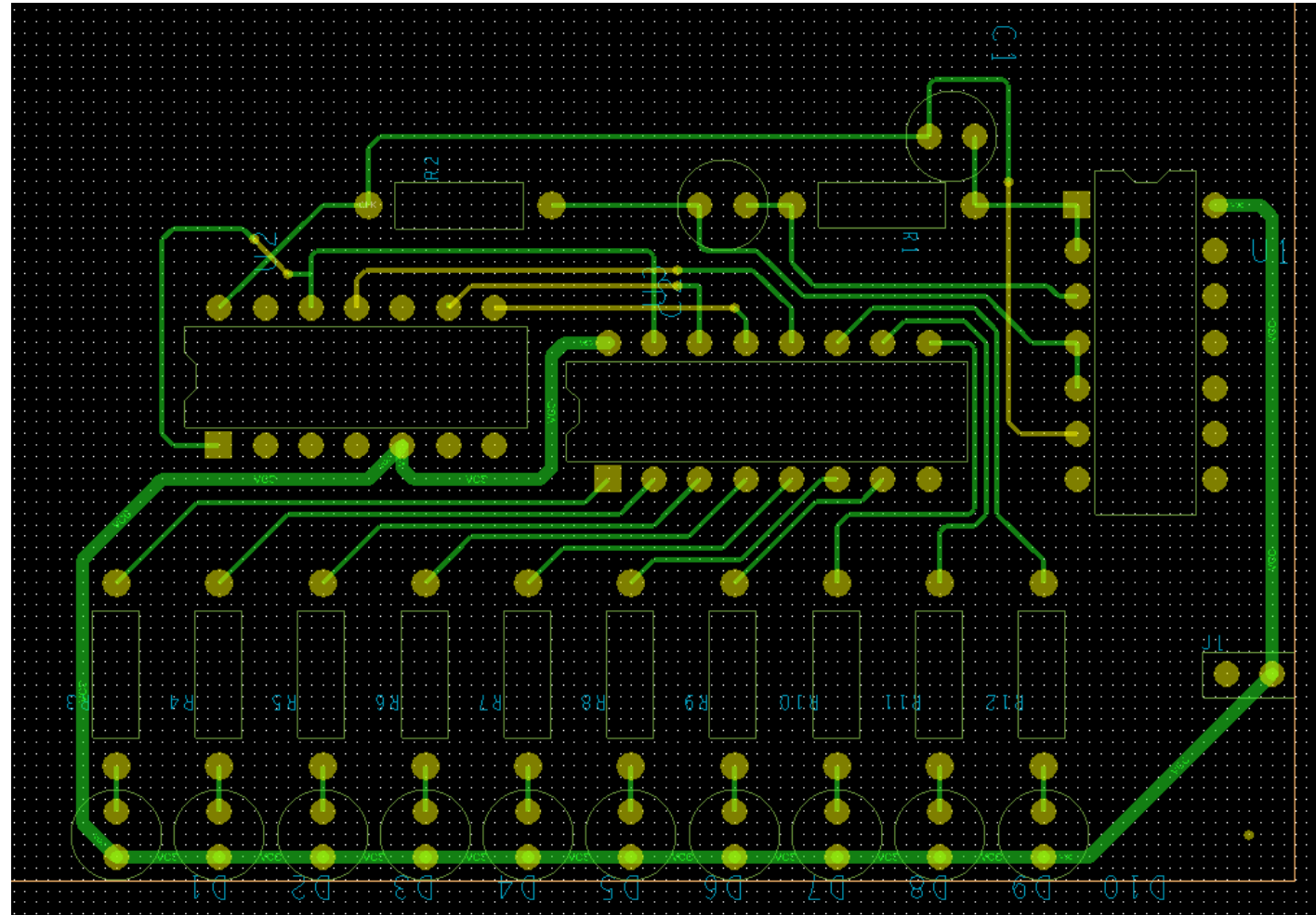
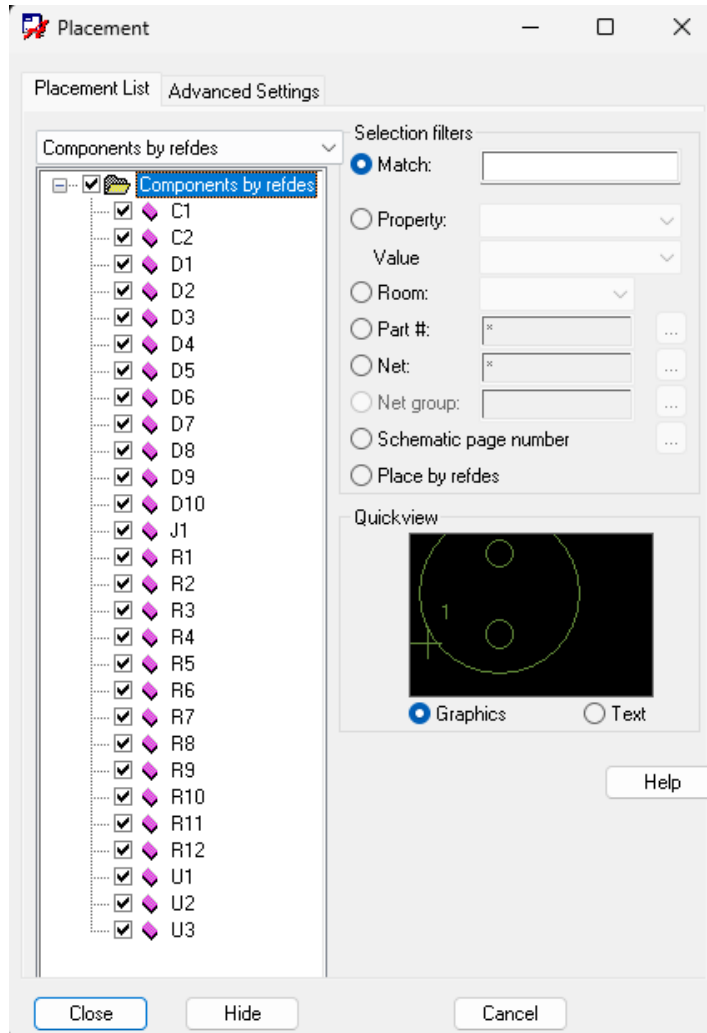
Properties

DRC

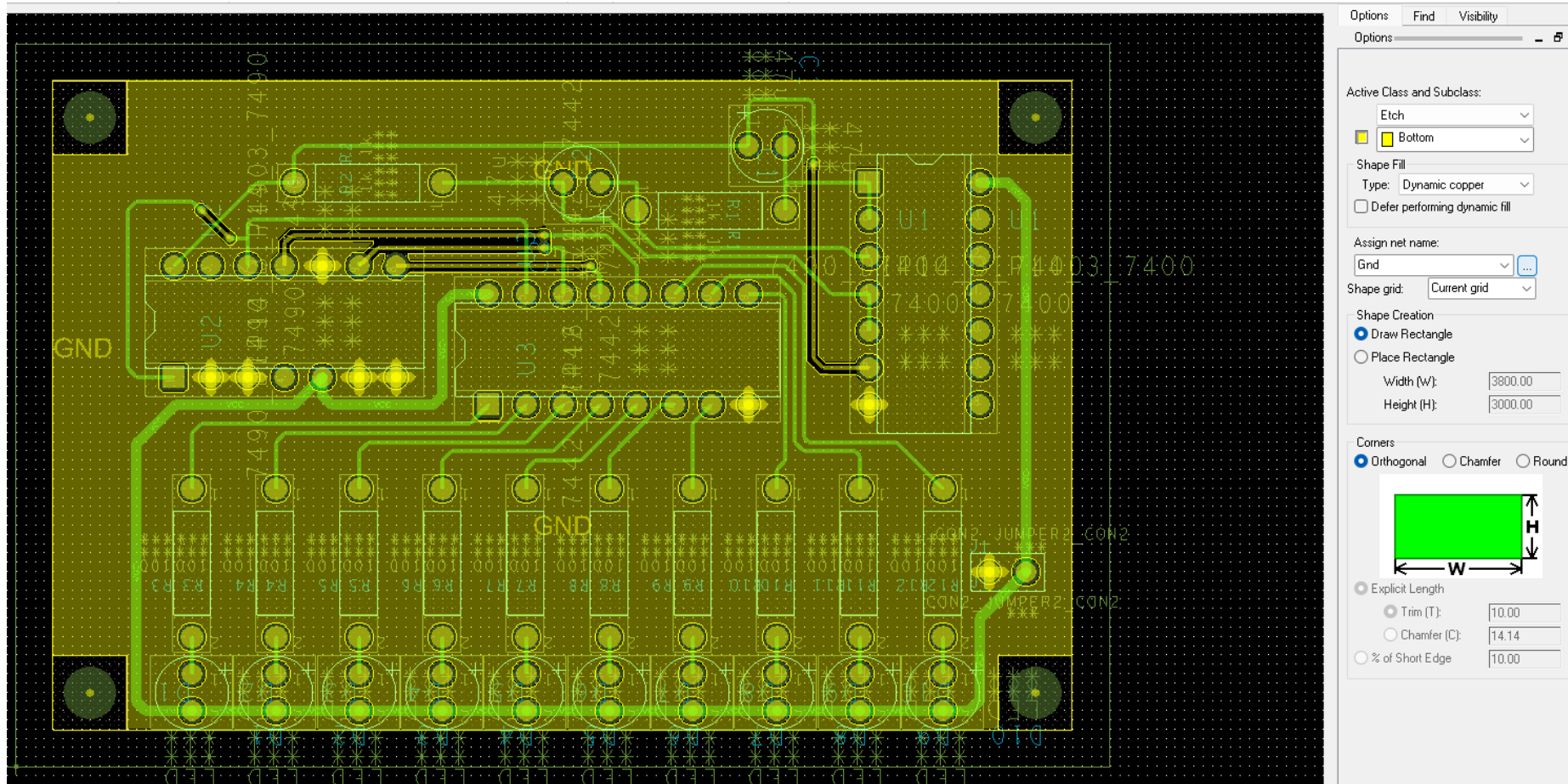
Idle DRC Sync on.

3. Layout

Part Placement and Routing



Ground Copper Pour Placement



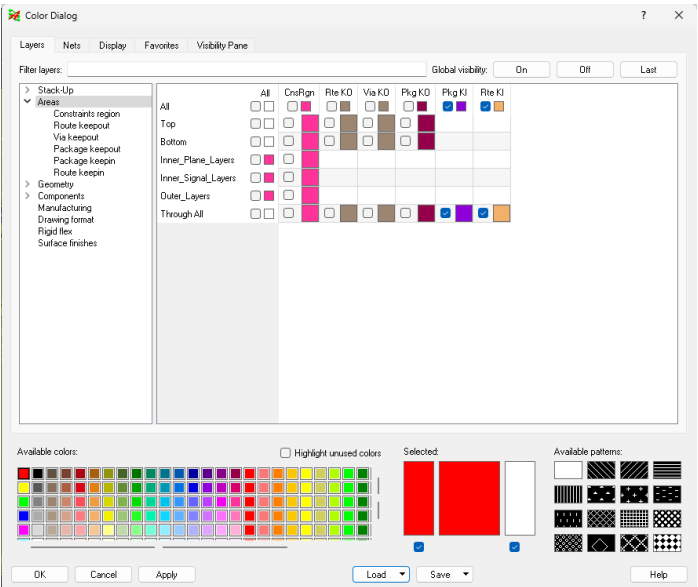
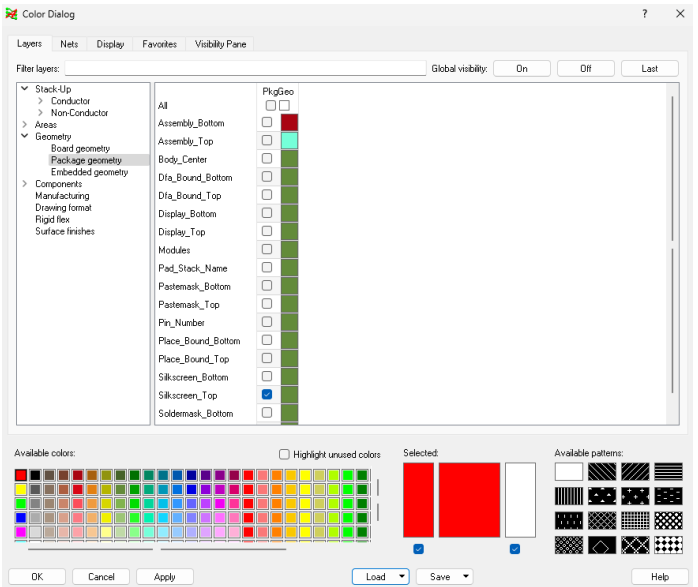
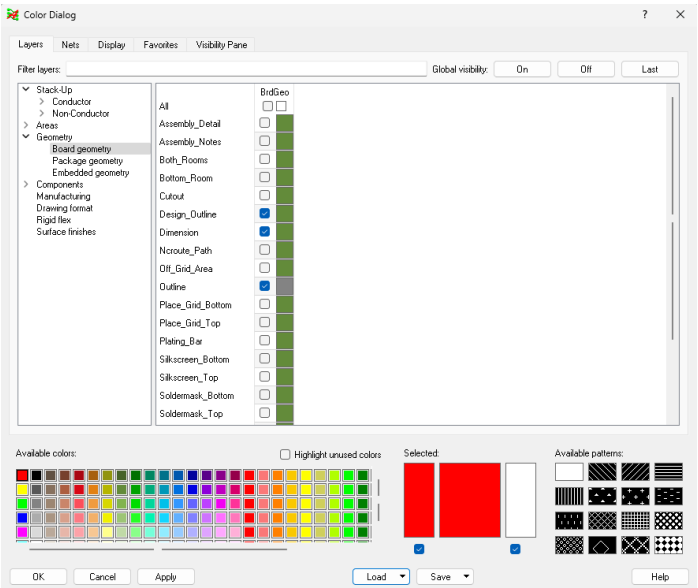
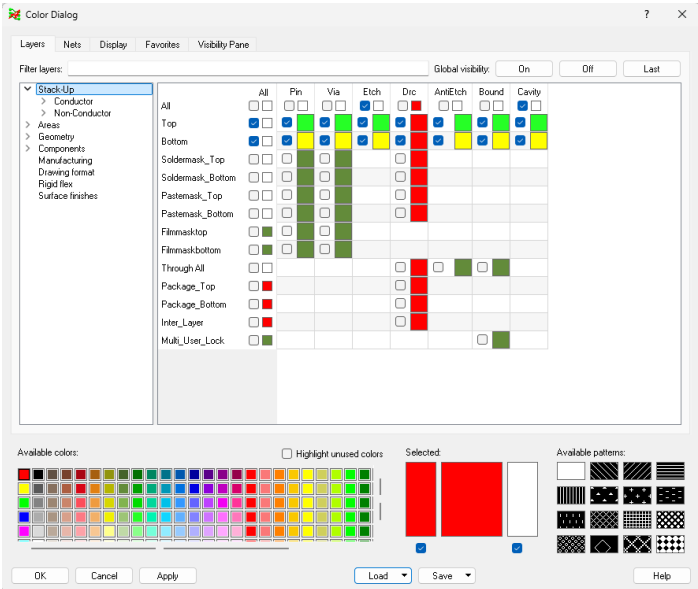
Design Rule Check (DRC)

The image displays a PCB layout on a green grid. The layout includes various components such as resistors (R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R42, R43, R44, R45, R46, R47, R48, R49, R50, R51, R52, R53, R54, R55, R56, R57, R58, R59, R60, R61, R62, R63, R64, R65, R66, R67, R68, R69, R70, R71, R72, R73, R74, R75, R76, R77, R78, R79, R80, R81, R82, R83, R84, R85, R86, R87, R88, R89, R90, R91, R92, R93, R94, R95, R96, R97, R98, R99, R100), capacitors (C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C42, C43, C44, C45, C46, C47, C48, C49, C50, C51, C52, C53, C54, C55, C56, C57, C58, C59, C60, C61, C62, C63, C64, C65, C66, C67, C68, C69, C70, C71, C72, C73, C74, C75, C76, C77, C78, C79, C80, C81, C82, C83, C84, C85, C86, C87, C88, C89, C90, C91, C92, C93, C94, C95, C96, C97, C98, C99, C100), and integrated circuits (U1, U2, U3, U4, U5, U6, U7, U8, U9, U10, U11, U12, U13, U14, U15, U16, U17, U18, U19, U20, U21, U22, U23, U24, U25, U26, U27, U28, U29, U30, U31, U32, U33, U34, U35, U36, U37, U38, U39, U40, U41, U42, U43, U44, U45, U46, U47, U48, U49, U50, U51, U52, U53, U54, U55, U56, U57, U58, U59, U60, U61, U62, U63, U64, U65, U66, U67, U68, U69, U70, U71, U72, U73, U74, U75, U76, U77, U78, U79, U80, U81, U82, U83, U84, U85, U86, U87, U88, U89, U90, U91, U92, U93, U94, U95, U96, U97, U98, U99, U100). The layout is connected by green traces. A 'GND' label is visible on the left side of the board.

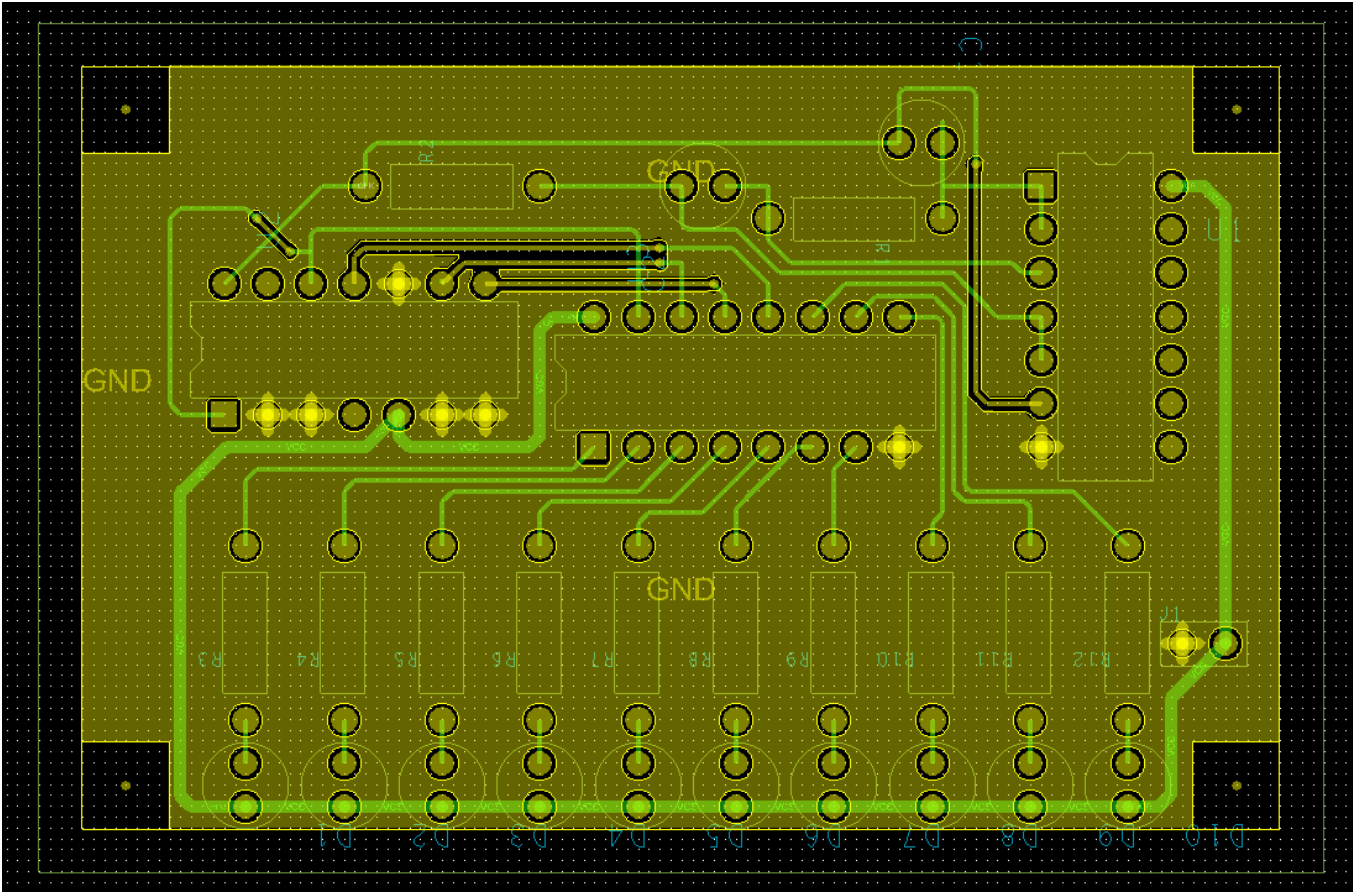
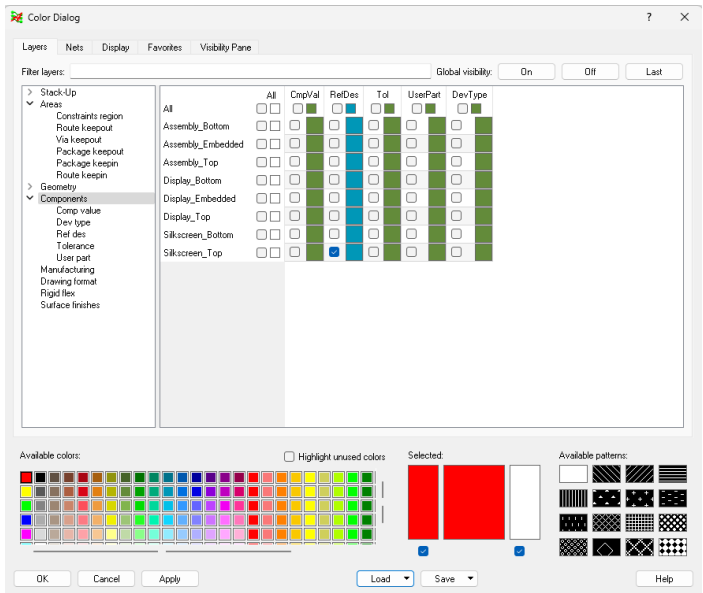
The 'Status' dialog box is open, showing the following information:

- Status**
- Symbols and nets**
 - Unplaced symbols: 0/28 0%
 - Unrouted nets: 0/30 0%
 - Unrouted connections: 0/56 0%
- Shapes**
 - Isolated shapes: 0
 - Unassigned shapes: 0
 - Out of date shapes: 0/1 [Update to Smooth](#)
- Dynamic fill:** ☒ Smooth ☐ Rough ☐ Disabled
- DRCs and Backdrills**
 - DRC errors: Up To Date 0 [Update DRC](#)
 - Shorting errors: 0 ☒ On-line DRC
 - Waived DRC errors: 0
 - Waived shorting errors: 0
- ☐ Out of date backdrills [Update Backdrill](#)
- Statistics**
 - Last saved by: 57
 - Editing time: 19 hours 14 minutes [Reset](#)
- [OK](#) [Refresh](#) [Help](#)
- DRC done.

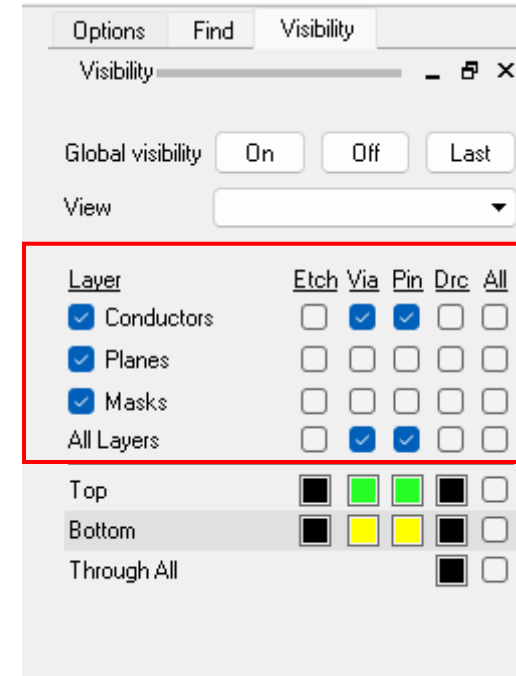
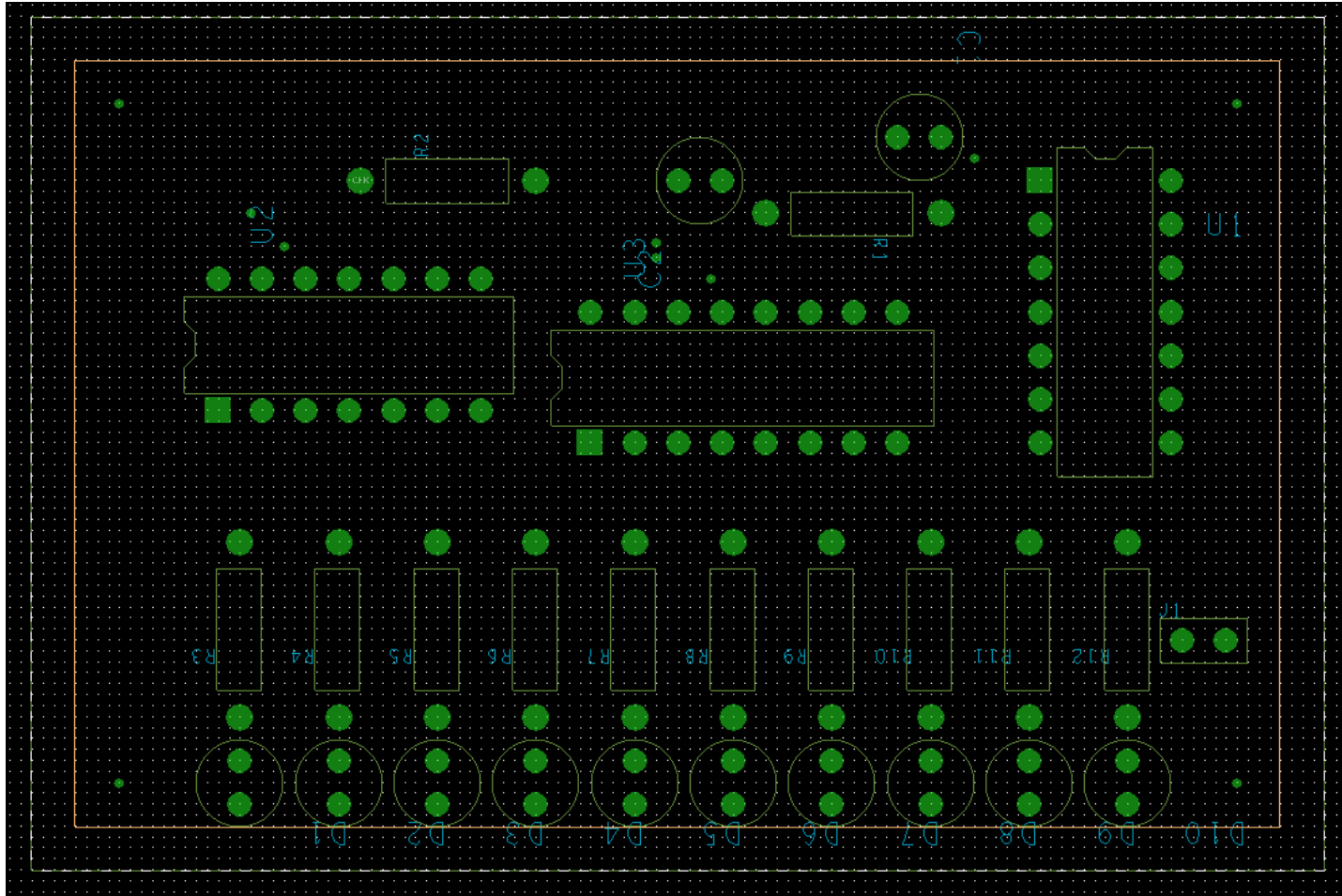
Color Setting



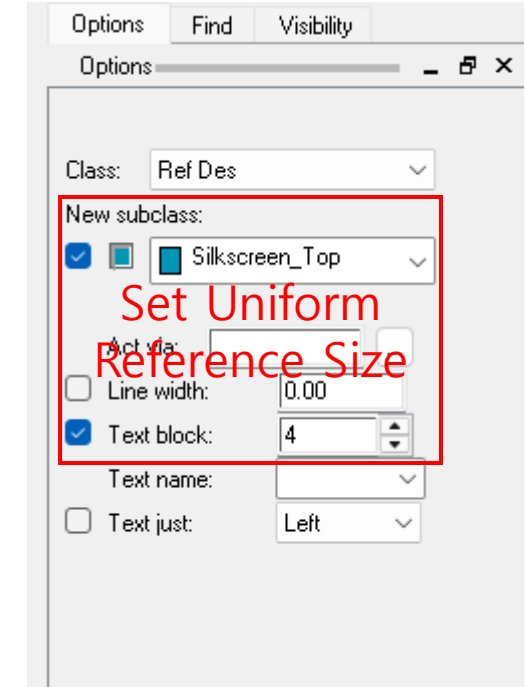
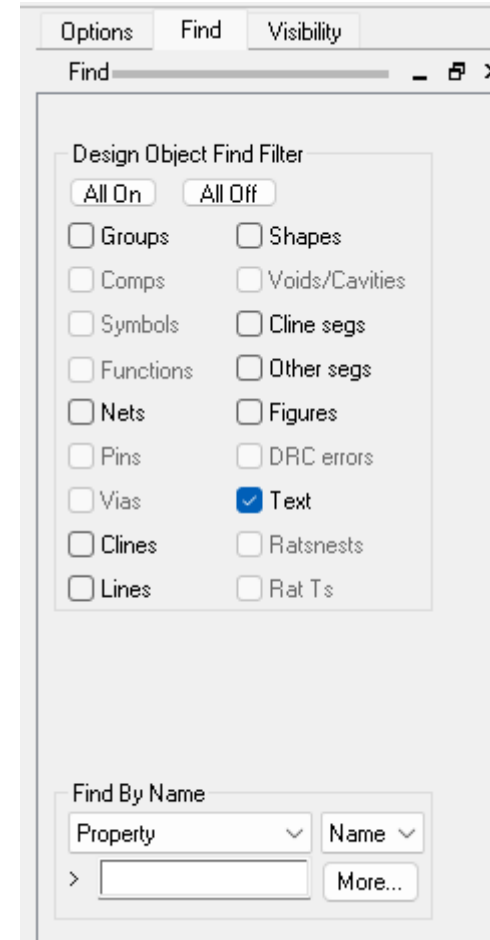
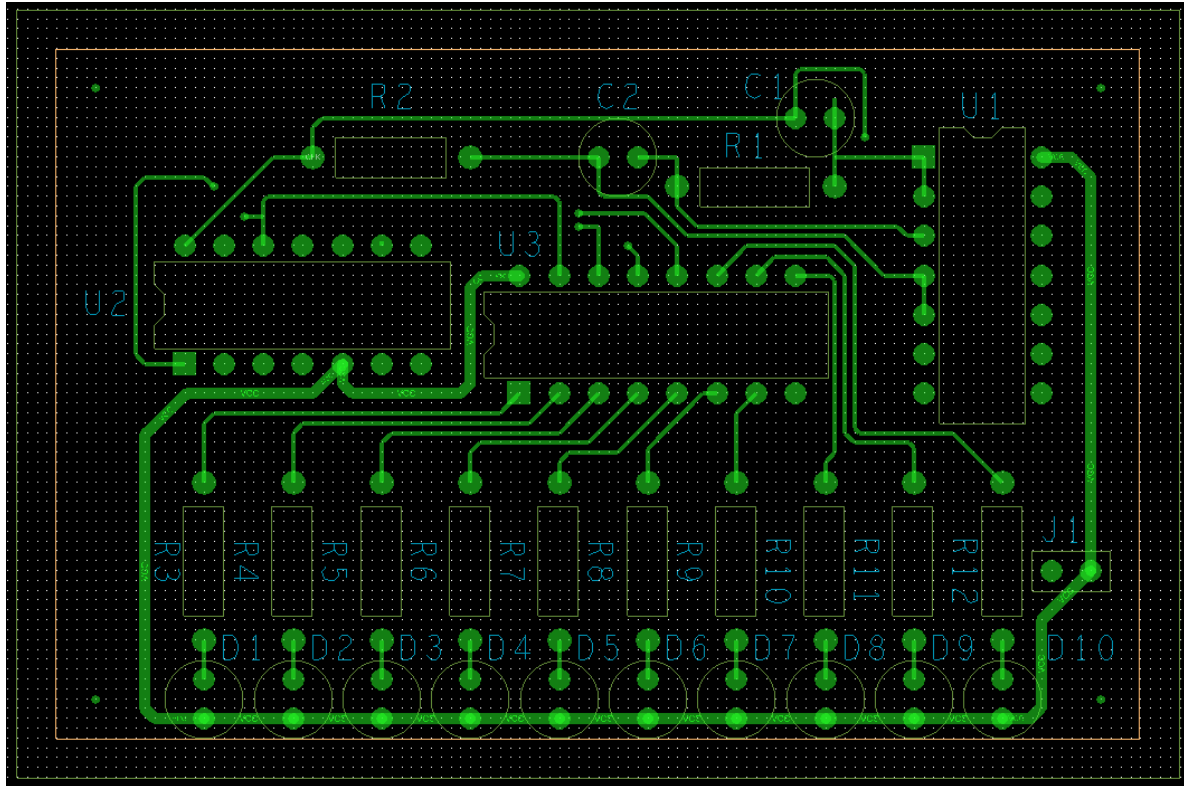
Color Setting



Reference Annotation

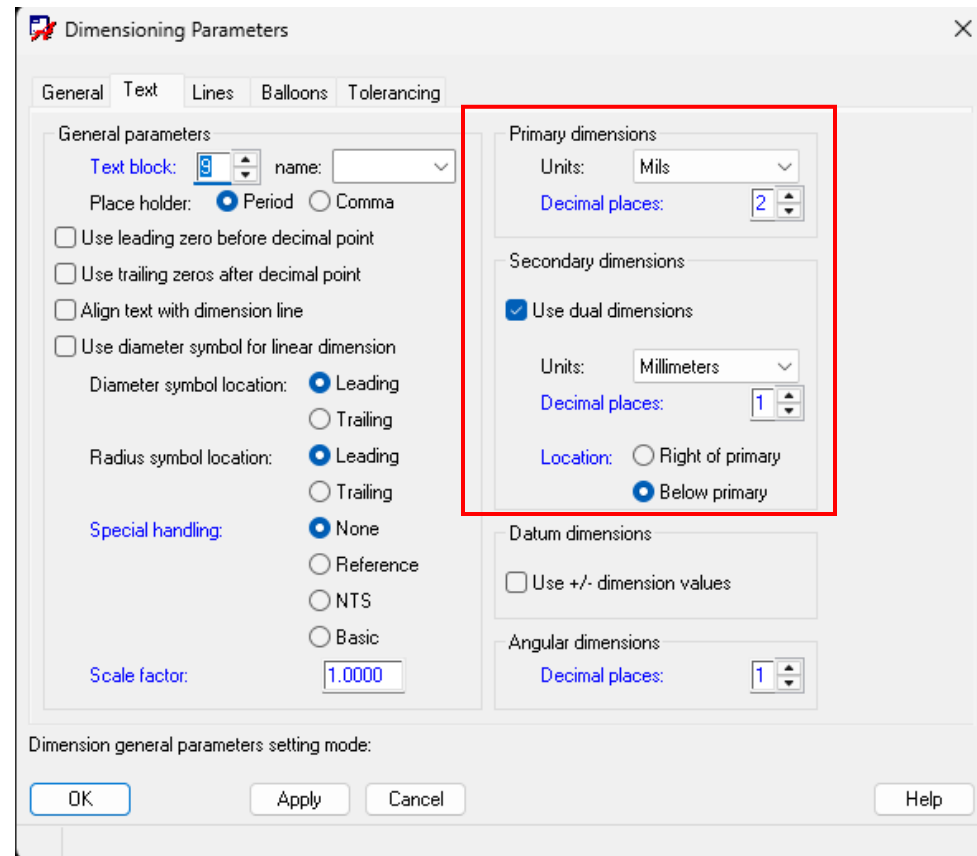
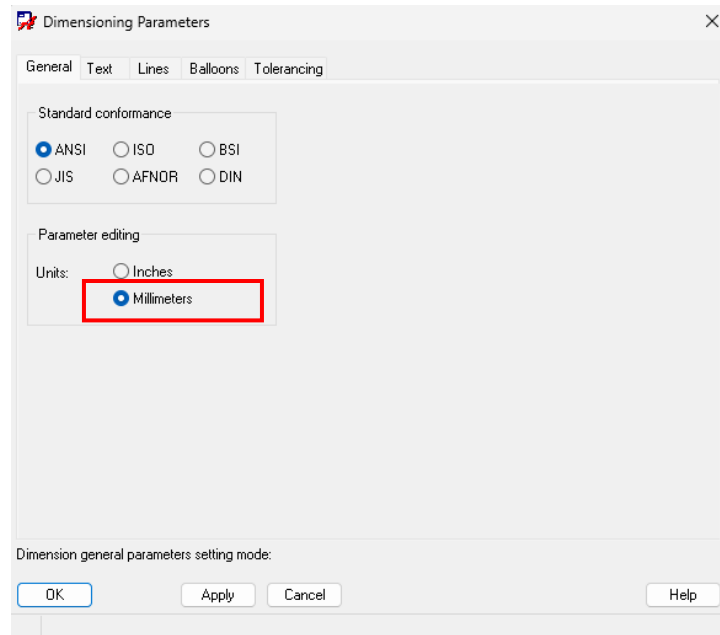


Reference Annotation

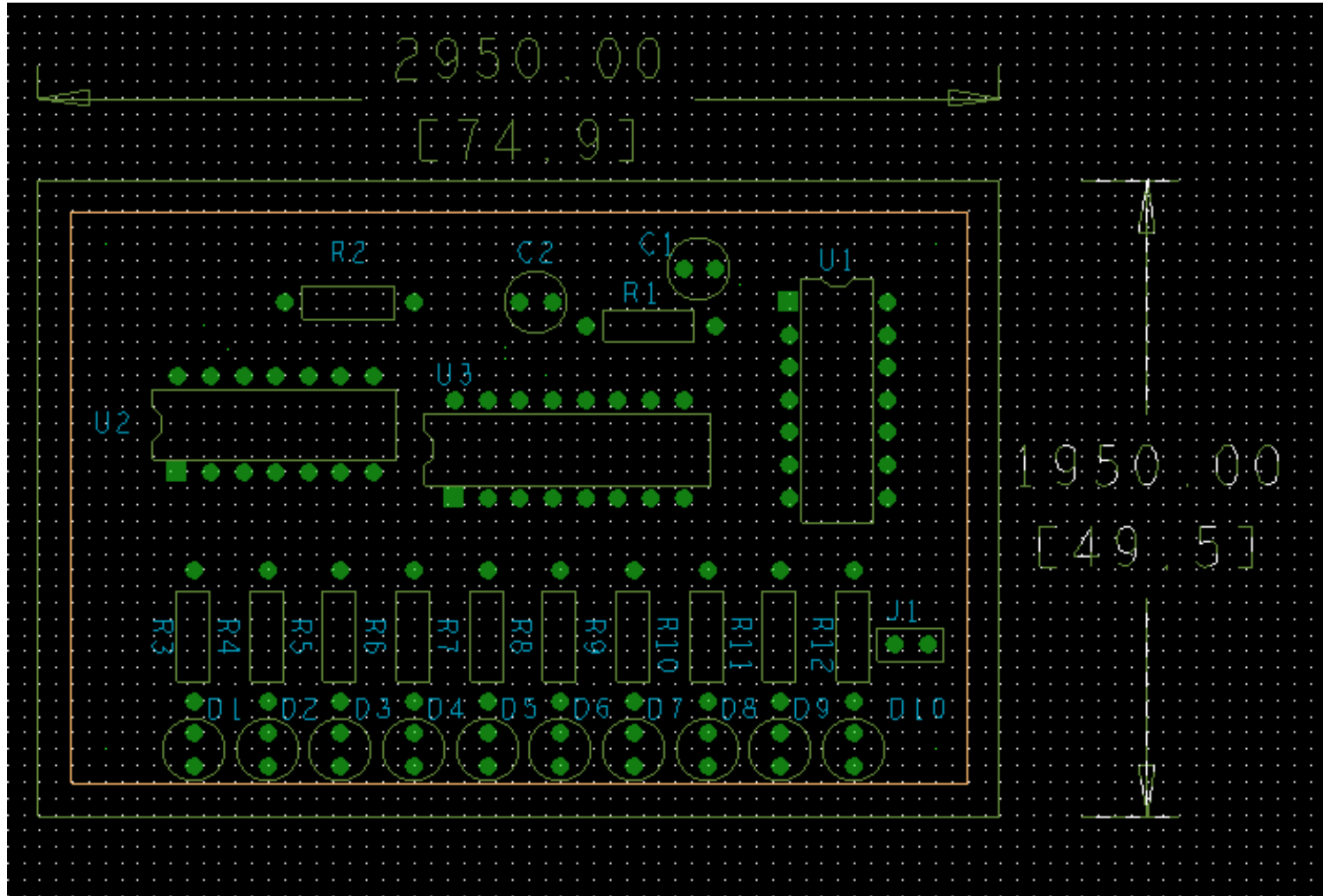


4. Drafting & Dimensions

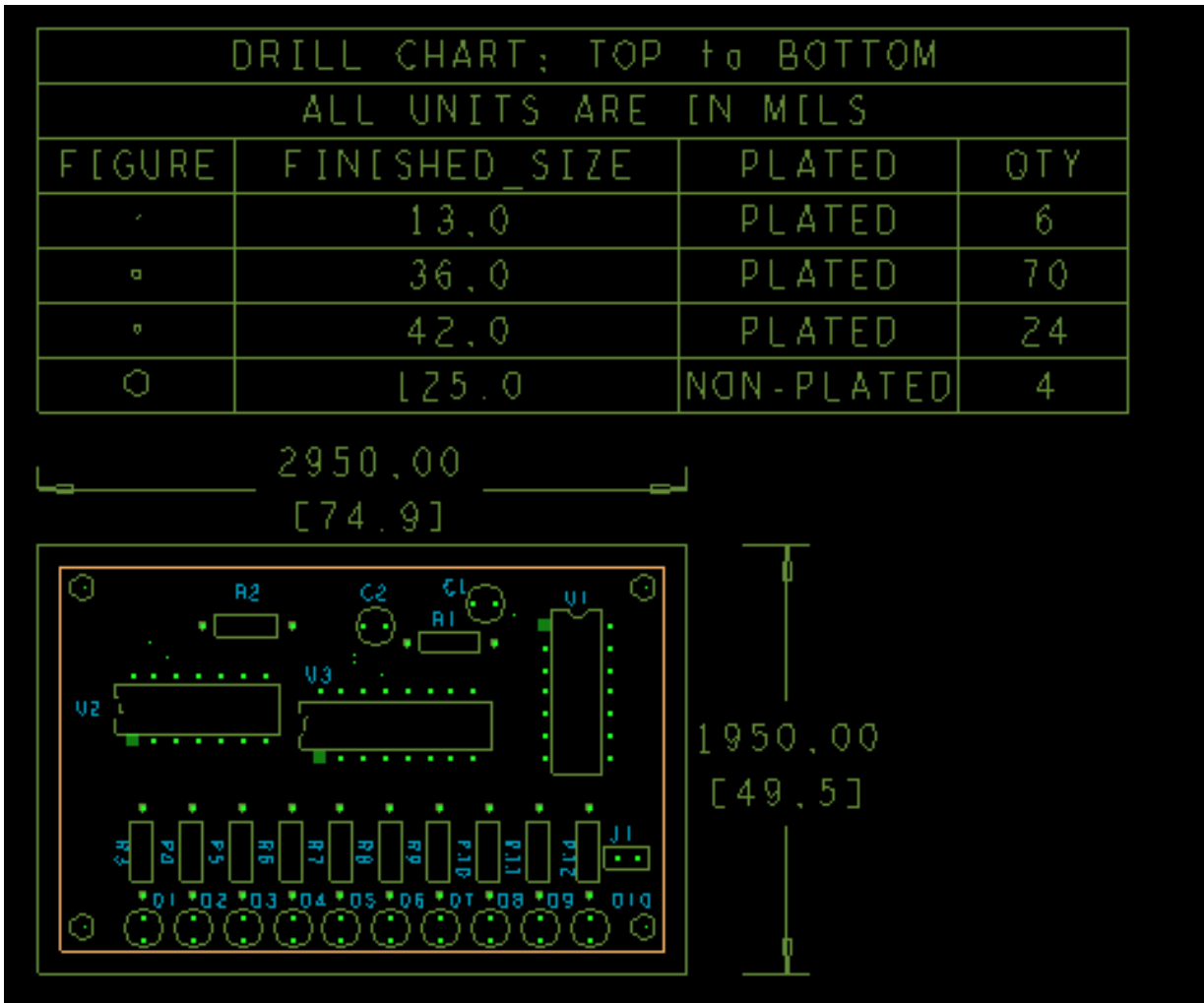
Drafting Dimension Settings



Add Dimensions



Add Drill Legend Table



Drill Customization

Drill/Slot Holes

#	Type	Size X	Size Y	Tool Size	+ Tolerance	- Tolerance	Symbol Figure	Symbol Characters	Symbol Size X	Symbol Size Y	Plating	Non-standard Drill	Quantity
1	Circle Drill	13.00			0.00	0.00	Cross		13.00	13.00	Plated		6
2	Circle Drill	36.00			0.00	0.00	Square		36.00	36.00	Plated		70
3	Circle Drill	42.00			0.00	0.00	Hexagon		42.00	42.00	Plated		24
4	Circle Drill	125.00			0.00	0.00	Hexagon		125.00	125.00	Non-Plated		4

Validate Merge Reset to design Reset to library Auto generate symbols Write report file CSV HTML Total quantity: 104

OK Cancel Help Library drill report

Drill Legend

Template file: default.mil.d Browse... Library...

Output unit: Mils

Legend title: DRILL CHART: \$lay_nams\$

C-Bore: COUNTERBORE/COUNTERSINK: \$lay_nams\$

Hole sorting method:
By hole size: ☒ Ascending ☐ Descending
By plating status: ☒ Plated first ☐ Non-plated first

Legends:
☒ Layer pair ☐ By layer ☐ Include C-Bore

Other Options:
Drill Legend Columns:
☒ Tolerance drill ☒ Tolerance travel
☒ Tool size ☒ Rotation ☒ Non-standard type
☐ Display total slot/drill count
☐ Separate slots from drills
☒ Suppress tolerance column if all values are 0's
☒ Suppress tool size column if all values are empty
☒ Suppress rotation column if all values are 0's

OK Cancel Help

Design Parameter Editor

Display Design Text Shapes Route Mfg Applications

Command parameters

Display:
Connect point size: 10.00
DRC marker size: 25.00
Rat T (Virtual pin) size: 35.00
Max island count: 500
Ratnest geometry: Joggled
Ratnest points: Closest endpoint

Display net names (OpenGL only):
☒ Clines
☒ Shapes
☒ Pins

Enhanced display modes:
☐ Plated holes
☐ Backdrill holes
☐ Non-plated holes
☐ Padless holes
☒ Connect points
☒ Filled pads
☒ Connect line endpoints
☐ Thermal pads
☐ Bus rats
☐ Waived DRCs
☐ Drill labels
☐ Design origin
☐ Drillpair driver pins
☐ Use secondary step models in 3D viewer

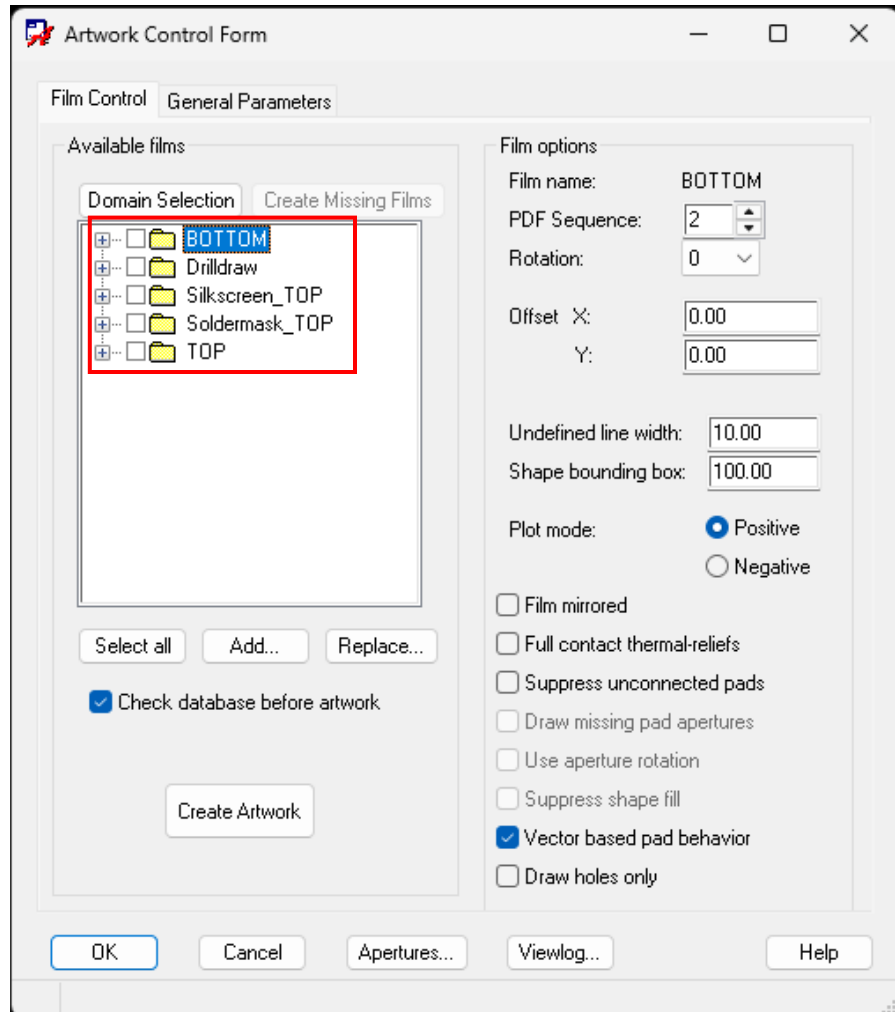
Grids:
☒ Grids on Setup grids ...

Parameter description:
Displays a pop-up menu that allows the specification of the closest distance between existing etch or pads (Closest endpoint) or between two pins (Pin to pin). The default is Closest endpoint.

OK Cancel Apply Help

5. Manufacturing Outputs

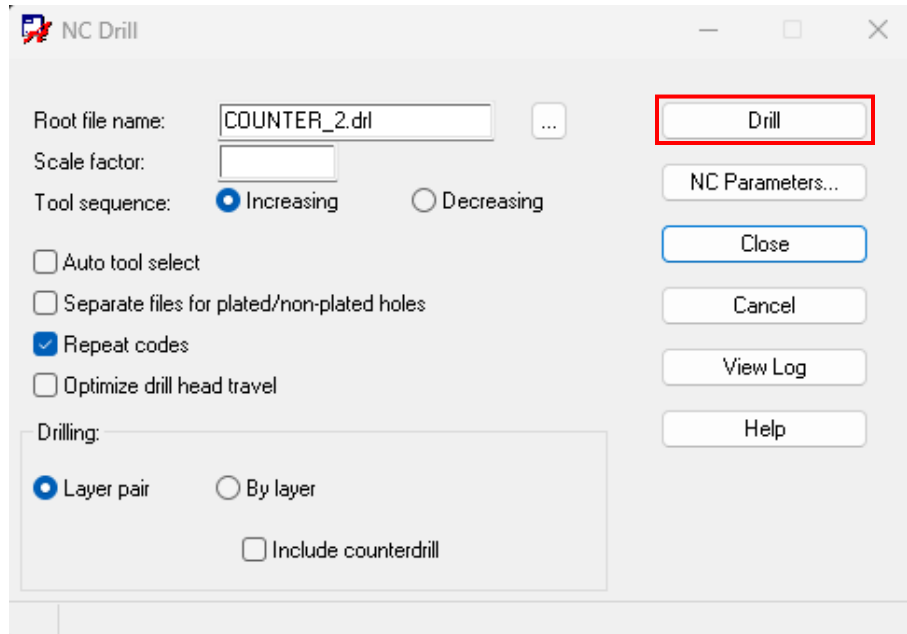
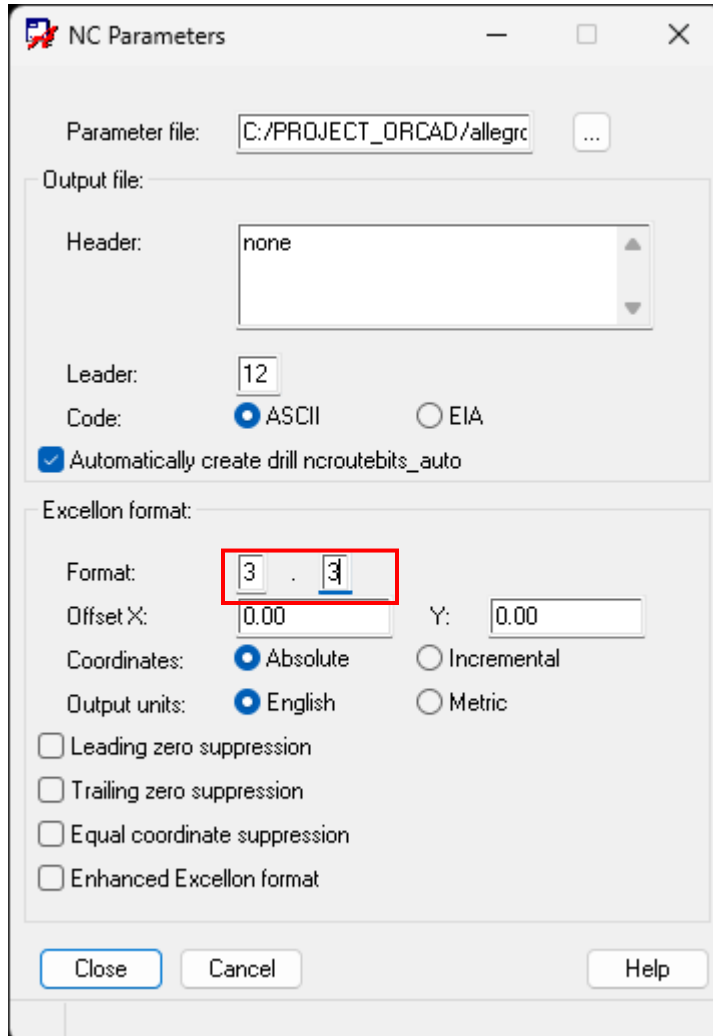
Generate Gerber Files



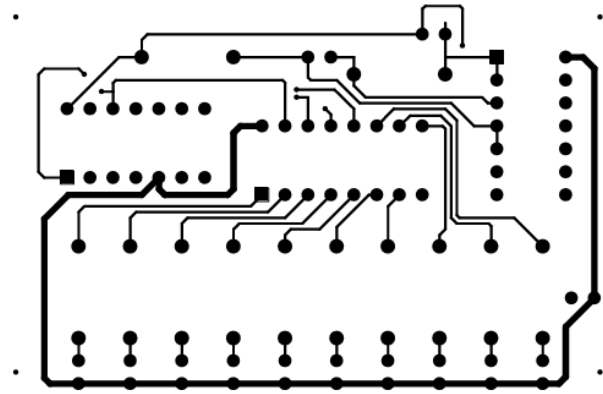
Generate Gerber Files

- TOP, BOTTOM, Soldermask_TOP, Silkscreen_TOP, Drilldraw

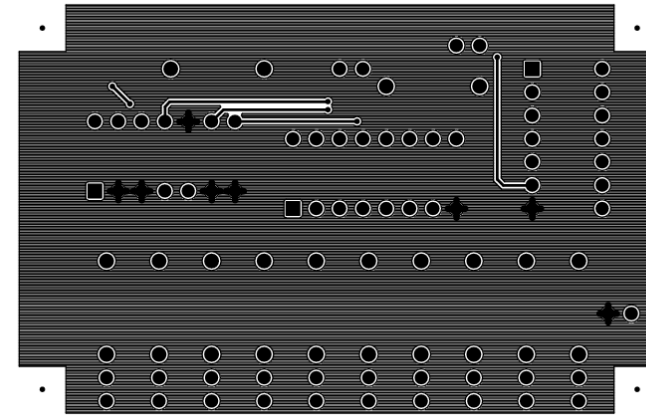
Generate NC Drill



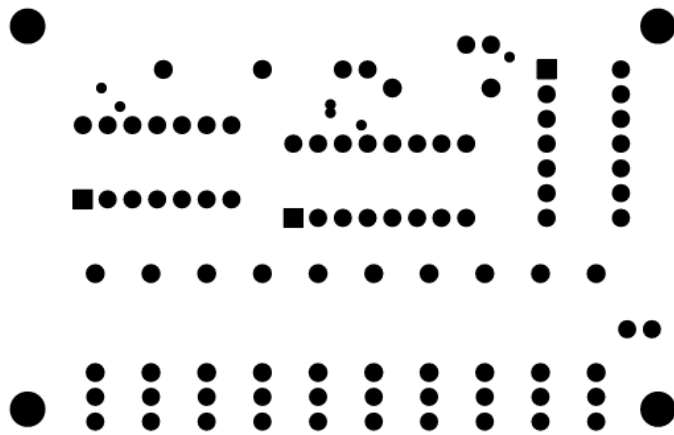
PCB Gerber Files



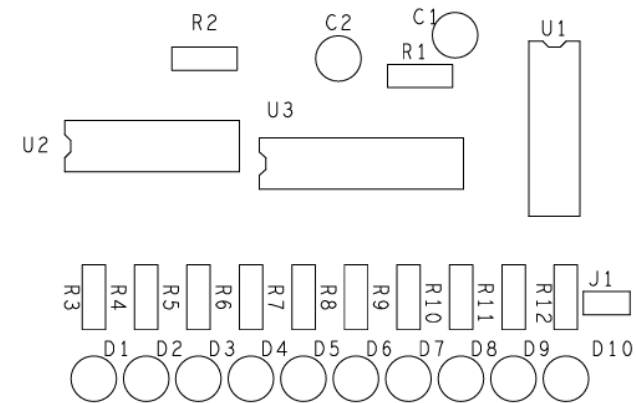
<Top Layer Pattern>



<Bottom Copper Layer>



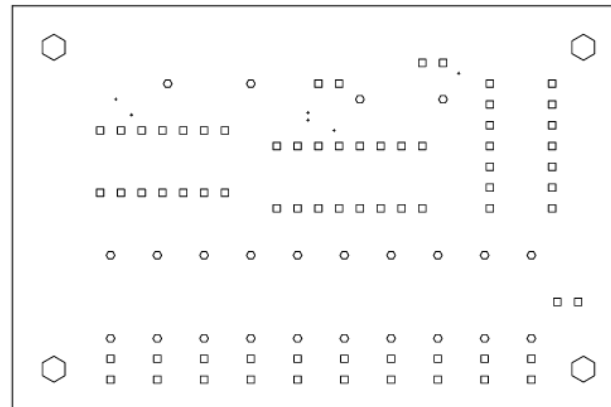
<Top Silkscreen>



<Top Soldermask>

PCB Gerber Files

DRILL CHART: TOP to BOTTOM			
ALL UNITS ARE IN MILS			
FIGURE	FINISHED_SIZE	PLATED	QTY
.	13.0	PLATED	6
□	36.0	PLATED	70
○	42.0	PLATED	24
⬡	125.0	NON-PLATED	4



<Drilldraw>

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